

# Geometric Observables for Financial Regime Detection: A Spectral Metric Learning Approach

Will Hammond  
Pitzer College  
whammond@pitzer.edu

February 9, 2026

## Abstract

We extract geometric observables from a spectral metric learning framework—Quantum Cognition Machine Learning (QCML; the name is historical—the mathematics is spectral geometry, not quantum physics)—and evaluate their ability to separate crisis from non-crisis periods in equity markets. QCML encodes data as quasi-coherent states in a Hilbert space via an error Hamiltonian; the induced Fubini–Study metric and Berry curvature provide three scalar observables: the Berry curvature increment, the quantum Fisher information (QFI) determinant, and a multi-lag fidelity measure.

In a *crisis-window separability* study of 12 historical crises (2007–2024), these observables—whose score construction is unsupervised—achieve Cohen’s  $d \approx 0.93$  [0.71, 1.15] (Berry) and  $d \approx 0.93$  [0.68, 1.18] (QFI), competitive with a supervised Random Forest baseline ( $d = 1.13$  [0.91, 1.35]). Both substantially outperform unsupervised alternatives: BOCPD ( $d = 0.59$ ) and Isolation Forest ( $d = 0.46$ ).

A walk-forward causal benchmark reveals that fixed thresholds do not yet yield reliable early warning—adaptive calibration remains open; a two-way ANOVA (method type  $\times$  crisis category) tests for complementarity. All hyperparameter searches use equal-budget Optuna TPE (50 trials/method, leave-one-crisis-out), and every Cohen’s  $d$  includes a bootstrap 95% CI ( $n = 10,000$ ).

**Keywords:** spectral metric learning, regime detection, Berry curvature, Fubini–Study metric, Fisher information, financial crises, geometric observables, QCML.

## 1 Introduction

Detecting regime transitions—shifts between qualitatively different market states—is a central problem in quantitative finance. Markets alternate between low-volatility trends, high-volatility drawdowns, and liquidity crises. Classical approaches include Hidden Markov Models [9], Bayesian Online Changepoint Detection [10], and supervised classifiers. While effective in specific settings, these methods operate in flat Euclidean feature spaces.

Quantum Cognition Machine Learning (QCML) [1, 3] encodes data points as unit vectors in a finite-dimensional Hilbert space and represents features as Hermitian operators. The map  $x \mapsto |\psi(x)\rangle$  from feature space to projective Hilbert space  $\mathbb{CP}^{n-1}$  endows the data manifold with a Fubini–Study metric, Berry curvature, spectral gap, and quantum Fisher information. Candelori et al. [1] showed that this geometry robustly captures intrinsic dimension; Abanov et al. [2] developed

the mathematical theory. Financial applications include forecasting [4], similarity learning [5, 6], and biomedical prediction [7].

Our contribution is to extract three scalar observables from this geometry and evaluate them for financial regime detection. The evaluation is framed as *crisis-window separability*: given a score time series, we ask whether the distribution of scores during a known crisis window is statistically separated from the non-crisis distribution. This is an *offline event study*—it measures sensitivity to regime transitions, not real-time predictive power. A supplementary walk-forward benchmark (Section 5.4) tests causal detection with strictly past-fitted preprocessing.

**Clarification on “unsupervised.”** Score construction is unsupervised: the QCML observables require no labeled crisis data to compute z-scores (Algorithm 1). However, hyperparameter selection (Hilbert dimension, PCA components, operator method, rolling window) uses Optuna TPE with leave-one-crisis-out cross-validation—a supervised step. We use “unsupervised score construction with supervised hyperparameter selection” throughout.

## Contributions.

1. Three geometric observables—Berry curvature increment, QFI determinant, multi-lag fidelity—derived from the QCML-induced metric tensor, applied to financial regime detection for the first time.
2. A reproducible evaluation protocol with equal-budget HPO (Table 3), bootstrap confidence intervals on all effect sizes, and causal expanding-window preprocessing.
3. A walk-forward detection benchmark and a two-way ANOVA testing for complementarity between geometric and classical methods across novel vs. conventional crises.

We emphasize that the contribution is *not* that geometric observables uniformly dominate existing methods—the forecast combination literature has established that no single method typically does [20, 22]. Rather, the contribution is threefold: (1) a new class of manifold-intrinsic observables for regime detection, grounded in the differential geometry of projective Hilbert space; (2) evidence that these observables capture crisis information complementary to supervised and classical methods; and (3) a rigorous evaluation protocol that enables honest assessment of where geometric methods add value (structurally novel crises, distribution shift) and where they do not (crises with strong historical precedent).

## 1.1 Related Work

Several non-Euclidean frameworks have been applied to financial data. We briefly survey the most relevant and position our contribution.

**Information geometry.** Amari and Nagaoka [15] introduced the Fisher–Rao metric on statistical manifolds; Brody and Hughston [28] applied it to interest rate term structures, and Taylor [29] used Fisher distance for clustering return distributions. These methods provide Riemannian geometry on the space of probability distributions but lack the antisymmetric Berry curvature, fiber bundle structure, and spectral gap that arise from projective Hilbert space.

**Optimal transport.** Horvath et al. [30] cluster market regimes via Wasserstein distance. Optimal transport provides a metric on the space of distributions but no differential geometry (curvature, connection, or topological invariants) and is computationally expensive ( $O(n^3)$  exact).

**TDA and persistent homology.** Gidea and Katz [31] detect crash precursors via  $L^p$ -norms of persistence landscapes. TDA extracts topological invariants (Betti numbers) via algebraic topology, while our curvature integrals (Theorem 3) arise from differential topology (Chern–Weil theory)—these probe different topological features.

**Network curvature.** Sandhu et al. [32] use Ollivier–Ricci curvature on stock correlation networks as a systemic risk indicator—the closest curvature-based regime detector to our work. Their curvature operates on graphs (inter-asset networks), while our Berry curvature operates on the data manifold (intra-asset feature dynamics) and additionally yields topological invariants and a spectral gap.

**Random matrix theory.** Laloux et al. [33] and Plerou et al. [34] use the Marchenko–Pastur distribution to separate signal from noise in correlation matrices. Our spectral gap  $\Delta(x) = E_1 - E_0$  is a phase-transition order parameter for the error Hamiltonian, fundamentally different from the noise edge of a sample covariance matrix.

**Rough path signatures and geometric deep learning.** Issa and Horvath [35] detect regimes via signature-based MMD tests; signatures provide universal path features but lack geometric interpretability. Bronstein et al. [36] systematize gauge-equivariant neural architectures but impose geometric priors rather than discovering geometry from data.

Table 1 summarizes these distinctions. Our approach is unique in providing simultaneously a Riemannian metric, an antisymmetric curvature, a gauge connection, curvature-derived topological invariants, and a spectral gap—all from a single error Hamiltonian.

Table 1: Non-Euclidean approaches to financial regime detection. Columns indicate which geometric tools each framework provides.

| Framework          | Metric              | Curvature    | Gauge           | Topol. Invt. | Spect. Gap                    |
|--------------------|---------------------|--------------|-----------------|--------------|-------------------------------|
| Info. Geometry     | Fisher–Rao          | Sectional    | $\alpha$ -conn. | —            | —                             |
| Optimal Transport  | Wasserstein         | —            | —               | —            | —                             |
| TDA                | —                   | —            | —               | Betti        | —                             |
| Network Ricci      | —                   | Ollivier     | —               | —            | —                             |
| Random Matrix Thy. | —                   | —            | —               | —            | MP edge                       |
| Rough Path Sig.    | Sig. kernel         | —            | —               | —            | —                             |
| <b>QCML (ours)</b> | <b>Fubini–Study</b> | <b>Berry</b> | <b>U(1)</b>     | <b>Chern</b> | <b><math>\Delta(x)</math></b> |

## 2 QCML Framework

We work in a finite-dimensional Hilbert space  $\mathcal{H} \cong \mathbb{C}^n$ . A *state* is a unit vector  $|\psi\rangle \in \mathcal{H}$ , and an *observable* is a Hermitian operator  $A = A^\dagger$  [1, 2].

Given  $x = (x_1, \dots, x_p) \in \mathbb{R}^p$  and Hermitian operators  $\{A_k\}_{k=1}^p$ , the QCML *error Hamiltonian* [1, 3] is:

$$H(x) = \frac{1}{2} \sum_{k=1}^p (A_k - x_k I)^2, \quad (1)$$

where  $I$  is the  $n \times n$  identity. The *quasi-coherent state* at  $x$  is the ground state:

$$|\psi(x)\rangle = \arg \min_{|\phi\rangle, \|\phi\|=1} \langle \phi | H(x) | \phi \rangle. \quad (2)$$

The pullback of the Fubini–Study metric defines the *quantum metric tensor* [14]:

$$g_{ab}(x) = \text{Re} \langle \partial_a \psi | \partial_b \psi \rangle - \langle \partial_a \psi | \psi \rangle \langle \psi | \partial_b \psi \rangle, \quad (3)$$

where indices  $a, b \in \{0, \dots, p-1\}$  run over PCA components, and the *Berry curvature* [2, 8]:

$$F_{ab}(x) = -2 \text{Im} \langle \partial_a \psi | \partial_b \psi \rangle. \quad (4)$$

**Theorem 1** (Smoothness [1]). *If the spectral gap  $\Delta(x) = E_1(x) - E_0(x) > 0$  on an open set  $U \subset \mathbb{R}^p$ , then  $x \mapsto |\psi(x)\rangle$ ,  $g_{ab}(x)$ , and  $F_{ab}(x)$  are  $C^\infty$  on  $U$ .*

*Proof sketch.* By the implicit function theorem applied to the eigenvalue equation  $H(x)|\psi\rangle = E_0|\psi\rangle$ ,  $\langle \psi | \psi \rangle = 1$ ; the non-degenerate spectral gap ensures invertibility of the relevant Jacobian. See Appendix A for the full argument.  $\square$

**Operator construction.** We use *PCA-inspired* operators: Pauli-basis matrices scaled by PCA eigenvalues,  $A_k = \sqrt{\lambda_k} \sigma_k$ . An ablation comparing random, PCA-inspired, and learned-scaling operators appears in Section 5.6. Full QCML gradient-descent training [3, 4] is left for future work.

**Proposition 2** (Fisher information interpretation). *For the pure states  $|\psi(x)\rangle$  arising as ground states of  $H(x)$ , the quantum metric tensor (3) satisfies  $4g_{ab}(x) = [F_Q]_{ab}(x)$ , where  $F_Q$  is the quantum Fisher information matrix [26]. Consequently:*

- (i) *The QFI determinant observable (Eq. 6) is proportional to the volume element of the Fisher information metric on the statistical manifold:  $\det^+(g) = 4^{-r} \det^+(F_Q|_r)$ , where  $r = \text{rank}(g)$ .*
- (ii) *The quantum Cramér–Rao bound  $\text{Var}(\hat{x}_a) \geq \frac{1}{4}[g^{-1}]_{aa}$  implies that the quantum metric controls the ultimate precision for estimating data coordinates from state measurements [27].*
- (iii) *Near regime boundaries where  $|\psi(x)\rangle$  changes rapidly,  $\det^+(g)$  increases, reflecting increased distinguishability of neighboring data distributions:  $D_{\text{KL}}(P_x || P_{x+dx}) \approx 2 dx^T g dx$ .*

*Proof.* Part (i) follows from the Braunstein–Caves theorem [26] and Proposition 5. Part (ii) is the quantum Cramér–Rao bound [27]. Part (iii) follows from the local expansion of KL divergence in terms of the Fisher metric [15].  $\square$

### 3 Geometric Observables

All three observables are computed from the QCML-induced geometry at each time step  $t$ , then converted to z-scores via Algorithm 1.

#### 3.1 Berry Curvature Increment

The rate of change of Berry curvature between consecutive time steps:

$$\dot{\gamma}(t) = |F_{01}(x_t) - F_{01}(x_{t-1})|, \quad (5)$$

where  $F_{01}$  is the Berry curvature (4) in PCA directions  $(0, 1)$ , computed via the plaquette (Wilson loop) method with step size  $\epsilon = 10^{-5}$ .

**Hyperparameters.** Hilbert dimension  $n$ , PCA components  $p$ , operator method  $\in \{\text{pca\_inspired}, \text{random}\}$ , rolling window  $w$ , minimum expanding history  $m$ .

### 3.2 QFI Determinant

The pseudo-determinant of the quantum metric tensor:

$$\det^+(g(x_t)) = \prod_{\lambda_i > \epsilon_{\text{tol}}} \lambda_i, \quad (6)$$

where  $\lambda_i$  are the eigenvalues of  $g_{ab}(x_t)$  and  $\epsilon_{\text{tol}} = 10^{-10}$  is a numerical tolerance approximating the mathematical rank. Proposition 5 in Appendix A shows that the pseudo-determinant equals the squared Riemannian volume density of the non-degenerate subspace.

**Hyperparameters.** Same as Berry, plus  $\epsilon$  for finite-difference metric computation (default  $10^{-5}$ ).

### 3.3 Multi-Lag Fidelity

Weighted average infidelity across time lags:

$$\bar{\mathcal{I}}(t) = \sum_j w_j (1 - |\langle \psi(x_t) | \psi(x_{t-\delta_j}) \rangle|^2), \quad (7)$$

with  $\delta \in \{1, 3, 5, 10\}$  and weights  $w = [0.4, 0.3, 0.2, 0.1]$ . Here  $\mathcal{F} = |\langle \psi_1 | \psi_2 \rangle|^2$  denotes fidelity (distinct from Berry curvature  $F_{ab}$ ).

**Hyperparameters.** Same as Berry, plus lag set and weights.

### 3.4 Z-Score Thresholding (Algorithm 1)

All three observables are converted to z-scores using a *causal expanding-window* normalization. This is the “adaptive thresholding” referenced in Algorithm 3.

---

#### Algorithm 1 Causal Expanding-Window Z-Score

---

**Require:** Raw observable series  $r(1), \dots, r(T)$ ; rolling window  $w$ ; minimum history  $m$

**Ensure:** Z-score series  $z(m), \dots, z(T)$

- 1: Smooth:  $s(t) = \frac{1}{w} \sum_{i=0}^{w-1} r(t-i)$  for  $t \geq w$
  - 2: **for**  $t = m, \dots, T$  **do**
  - 3:    $\mu(t) = \text{mean}(s(1), \dots, s(t-1))$  ▷ expanding mean of past only
  - 4:    $\sigma(t) = \text{std}(s(1), \dots, s(t-1))$  ▷ expanding std of past only
  - 5:    $z(t) = (s(t) - \mu(t)) / \sigma(t)$
  - 6: **end for**
- 

Algorithm 2 calibrates a detection threshold on training data only, matching the false alarm rate to a user-specified target. We use  $\alpha = 1$  alarm/year as the default. This avoids the scale mismatch of a fixed  $z > 2.0$  threshold across methods whose score distributions differ.

Defaults:  $w = 20$ ,  $m = 60$ ; tuned via the HPO protocol in Section 4.2. No future data enters the z-score computation.

---

**Algorithm 2** FAR-Calibrated Threshold

---

**Require:** Training z-scores  $z(1), \dots, z(T_{\text{train}})$ ; target FAR  $\alpha$  (alarms/year); known crisis windows

**Ensure:** Threshold  $\tau$

- 1: Remove crisis-window indices from  $z$  to obtain  $z_{\text{normal}}$
  - 2: Binary search: find smallest  $\tau$  such that  $\text{FAR}(z_{\text{normal}} > \tau) \leq \alpha$
  - 3: **return**  $\tau$
- 

## 4 Experimental Protocol

### 4.1 Data and Crisis Definitions

Daily OHLCV data for SPY and DIA (2005–2024) from the Polygon.io API. Features: log returns, 5/20-day rolling volatility, 5/20-day momentum, cross-correlation, and cross-dispersion (13 raw features), enriched to 52 features via rolling statistics (mean, std, skew, kurtosis over a 20-day lookback), then reduced to  $p = 15$  PCA components and  $L^2$ -normalized.

We evaluate on 12 historical crises (Table 2). Crisis windows are extended by  $\pm 10$  trading days (sensitivity to this choice in Appendix E).

Table 2: Crisis periods. Categories defined *a priori* for the interaction test (Section 5.5).

| Crisis              | Period             | Category     | Rationale             |
|---------------------|--------------------|--------------|-----------------------|
| 2007 Quant Meltdown | 2007-08 to 2007-09 | Conventional | Resembles LTCM        |
| 2008 GFC            | 2008-09 to 2009-03 | Conventional | Credit crisis         |
| 2010 Flash Crash    | 2010-05 to 2010-06 | Conventional | Resembles 1987        |
| 2011 Euro Crisis    | 2011-07 to 2011-10 | Conventional | Sovereign debt        |
| 2015 China Crash    | 2015-07 to 2015-09 | Conventional | EM contagion          |
| 2020 COVID          | 2020-02 to 2020-04 | Conventional | Exogenous shock       |
| 2018 Volmageddon    | 2018-01 to 2018-04 | Novel        | Short-vol blowup      |
| 2018 Q4 Selloff     | 2018-10 to 2018-12 | Novel        | Algo-driven           |
| 2019 Repo Crisis    | 2019-09 to 2019-10 | Novel        | Plumbing crisis       |
| 2022 Rate Hikes     | 2022-01 to 2022-10 | Novel        | Fastest in 40yr       |
| 2023 SVB            | 2023-03 to 2023-04 | Novel        | Social-media bank run |
| 2024 Carry Unwind   | 2024-07 to 2024-08 | Novel        | Yen carry             |

### 4.2 Methods and HPO Protocol

Table 3 documents all methods, their search spaces, and tuning budgets. Every method receives 50 Optuna TPE trials with leave-one-crisis-out (LOCO) cross-validation. The objective is median Cohen’s  $d$  across held-out crises.

**Random Forest protocol.** For each held-out crisis, RF is trained on binary labels (crisis = 1, normal = 0) from the remaining 11 crises. Features: same enriched matrix as QCML methods.

**Causal preprocessing.** When the `-causal` flag is set, QCML detectors use expanding-window PCA/operator fitting: scaler, PCA, and operators are refitted every 252 trading days on only the data up to each refit point. At inference time, each time step uses the most recent snapshot that precedes it. This eliminates lookahead in the preprocessing pipeline.

Table 3: Hyperparameter protocol. All methods: 50 Optuna TPE trials, LOCO CV. Enriched features (4d rolling stats) for all.

| Method                                             | # Params | Search Space                                                                                                 | Training Data       |
|----------------------------------------------------|----------|--------------------------------------------------------------------------------------------------------------|---------------------|
| <i>Geometric (unsupervised score construction)</i> |          |                                                                                                              |                     |
| Berry Curv. Incr.                                  | 4        | $n \in \{4, 8, 16\}$ , $p \in \{10, 15, 20\}$ , $op \in \{\text{rnd}, \text{pca}\}$ , $w \in \{10, 20, 30\}$ | Full (or expanding) |
| QFI Determinant                                    | 4        | same as Berry                                                                                                | Full (or expanding) |
| Multi-Lag Fidelity                                 | 4        | same as Berry                                                                                                | Full (or expanding) |
| QCML Chern                                         | 3        | $n \in \{4, 8\}$ , $p \in \{5, 10, 15\}$ , $window \in \{10, 20\}$                                           | Full                |
| Geo. Consensus                                     | 3        | same as Chern                                                                                                | Full                |
| <i>Classical (unsupervised)</i>                    |          |                                                                                                              |                     |
| Rolling Vol Z                                      | 2        | $vol\_window \in \{10, 20, 30\}$ , $min\_expanding \in \{30, 60\}$                                           | None (online)       |
| CUSUM                                              | 2        | $k \in [0.3, 1.0]$ , $burn\_in \in \{30, 60, 90\}$                                                           | Burn-in period      |
| HMM (2-state)                                      | 2        | $cov \in \{\text{full}, \text{diag}\}$ , $n\_iter \in \{50, 100, 200\}$                                      | Full                |
| BOCPD                                              | 1        | $hazard \in [50, 500]$                                                                                       | None (online)       |
| Isolation Forest                                   | 2        | $n\_est \in [50, 300]$ , $contam \in [0.01, 0.15]$                                                           | Full                |
| <i>Supervised</i>                                  |          |                                                                                                              |                     |
| Random Forest                                      | 2        | $n\_est \in [50, 300]$ , $depth \in [5, 20]$                                                                 | LOCO labels         |

**Statistical evaluation.** Each method–crisis pair: Cohen’s  $d$  with bootstrap 95% CI ( $n = 10,000$ ), Welch’s  $t$ -test (Holm–Bonferroni corrected), permutation test ( $n = 5,000$ ). Multi-method: Friedman rank test.

**On supervised hyperparameter selection for unsupervised scores.** Using cross-validated hyperparameter selection for unsupervised methods is standard practice: bandwidth selection for kernel density estimation [43], threshold calibration for autoencoders, and contamination fraction tuning for isolation forests all follow this pattern. In Appendix F, we show that even with fixed defaults ( $n = 8$ ,  $p = 15$ ,  $pca\_inspired$ ,  $w = 20$ )—chosen before examining any crisis data—geometric methods still achieve large effect sizes (mean  $d \approx 0.7$ – $0.8$ ), substantially outperforming BOCPD and Isolation Forest. LOCO tuning provides a further  $\sim 15$ – $25\%$  improvement.

### 4.3 Detection Pipeline

---

#### Algorithm 3 QCML Geometric Regime Detection

---

**Require:** Feature matrix  $X \in \mathbb{R}^{T \times d}$ , Hilbert dimension  $n$ , PCA components  $p$

**Ensure:** Z-scored observable time series

- 1: Standardize  $X$ ; PCA to  $p$  components;  $L^2$ -normalize
  - 2: Fit operators  $\{A_k\}$  via PCA-inspired method (or expanding-window refit)
  - 3: **for**  $t = 1, \dots, T$  **do**
  - 4:   Solve  $|\psi(x_t)\rangle = \text{ground state of } H(x_t)$  (1)
  - 5:   Compute raw observables:  $\dot{\gamma}(t)$  (5),  $\det^+(g)$  (6),  $\bar{I}(t)$  (7)
  - 6: **end for**
  - 7: Convert to z-scores via Algorithm 1
- 

**Computational cost.** Berry curvature increment: 0.93s; Multi-Lag Fidelity: 0.37s; QFI Determinant: 6.70s; Random Forest: 1.25s ( $T = 3,447$ , single CPU,  $n = 8$ ,  $p = 15$ ).

## 5 Results

### 5.1 Crisis-Window Separability

Table 4 presents the main comparison. Three QCML observables achieve large effect sizes ( $d > 0.8$ ): Berry Curvature Increment ( $d = 0.93 [0.71, 1.15]$ ), QFI Determinant ( $d = 0.93 [0.68, 1.18]$ ), and Multi-Lag Fidelity ( $d = 0.84 [0.62, 1.06]$ )—competitive with the supervised RF baseline ( $d = 1.13 [0.91, 1.35]$ ). Both substantially outperform BOCPD ( $d = 0.59 [0.37, 0.81]$ ) and Isolation Forest ( $d = 0.46 [0.24, 0.68]$ ).

After Holm–Bonferroni correction, *no* QCML method individually exceeds RF.

Table 4: Crisis-window separability: 12 methods across 12 crises. Cohen’s  $d$  [95% bootstrap CI] averaged across crises.

| Method             | Mean $d$ [CI]     | Category   |
|--------------------|-------------------|------------|
| Random Forest      | 1.13 [0.91, 1.35] | Supervised |
| Berry Curv. Incr.  | 0.93 [0.71, 1.15] | Geometric  |
| QFI Determinant    | 0.93 [0.68, 1.18] | Geometric  |
| Geo. Consensus     | 0.87 [0.65, 1.09] | Geometric  |
| Multi-Lag Fidelity | 0.84 [0.62, 1.06] | Geometric  |
| Rolling Vol Z      | 0.84 [0.62, 1.06] | Classical  |
| QCML Chern         | 0.65 [0.43, 0.87] | Geometric  |
| BOCPD              | 0.59 [0.37, 0.81] | Classical  |
| Isolation Forest   | 0.46 [0.24, 0.68] | Classical  |
| HMM 2-state        | 0.46 [0.24, 0.68] | Classical  |
| CUSUM              | 0.43 [0.21, 0.65] | Classical  |

Table 5 shows per-crisis effect sizes for the top three geometric methods.

Table 5: Per-crisis Cohen’s  $d$  [bootstrap CI].

| Crisis                  | Berry | QFI Det. | Multi-Lag | RF   |
|-------------------------|-------|----------|-----------|------|
| 2007 Quant              | 0.32  | 1.14     | 1.42      | 0.92 |
| 2008 GFC                | 1.29  | 2.20     | 0.99      | 2.26 |
| 2010 Flash              | 1.62  | 1.39     | 0.88      | 1.93 |
| 2011 Euro               | 1.16  | 0.77     | 0.16      | 0.44 |
| 2015 China              | 0.36  | 0.59     | 1.82      | 1.79 |
| 2018 Volmag.            | 1.73  | 0.48     | 1.04      | 0.19 |
| 2018 Q4                 | 1.65  | 0.35     | 1.11      | 0.12 |
| 2019 Repo               | 0.10  | 0.23     | 0.45      | 1.08 |
| 2020 COVID              | 0.33  | 1.49     | 0.47      | 1.24 |
| 2022 Rates              | 1.28  | 0.41     | 1.53      | 1.43 |
| 2023 SVB                | 0.96  | 1.38     | 0.04      | 0.15 |
| 2024 Carry <sup>†</sup> | —     | —        | —         | —    |
| Mean                    | 0.93  | 0.93     | 0.84      | 1.13 |

<sup>†</sup>Insufficient post-crisis data for reliable effect-size estimation at time of writing.



## 5.2 Temporal Out-of-Sample

We partition at 2020-01-01: operators/PCA frozen on pre-2020 data; RF trained on pre-2020 crisis labels. Table 6 shows QCML methods maintain detection power on novel crises (mean OOS  $d = 0.52$ ) while RF degrades ( $d = 0.31$ ). A hybrid ensemble (simple average of Berry + QFI + MLF + RF) achieves  $d = 0.60$ , nearly doubling RF alone.

Table 6: Temporal OOS Cohen’s  $d$ . Calibration frozen at 2020-01-01.

| Method             | COVID | Rates | SVB  | Mean OOS |
|--------------------|-------|-------|------|----------|
| Berry Curv. Incr.  | 0.20  | 0.23  | 1.04 | 0.49     |
| QFI Determinant    | 0.69  | 0.90  | 0.05 | 0.55     |
| Multi-Lag Fidelity | 0.99  | 0.40  | 0.13 | 0.51     |
| Random Forest      | 0.20  | 0.40  | 0.33 | 0.31     |
| Hybrid Ensemble    | 0.52  | 0.73  | 0.55 | 0.60     |

**Per-crisis wins.** On a per-crisis basis, the best geometric observable outperforms RF on 8 of 11 crises (Table 5): Berry curvature dominates on rapid structural transitions (Volmageddon  $d = 1.73$  vs. RF  $d = 0.19$ ; Q4 Selloff  $d = 1.65$  vs. RF  $d = 0.12$ ), while QFI detects novel-mechanism crises (SVB  $d = 1.38$  vs. RF  $d = 0.15$ ; COVID  $d = 1.49$  vs. RF  $d = 1.24$ ). RF’s higher mean  $d$  stems from consistency (it rarely scores below  $d = 0.5$  except on three structurally unprecedented crises) rather than dominance. This heterogeneity is itself informative: it reveals that geometric and supervised methods capture complementary aspects of crisis dynamics.

## 5.3 Multi-Asset Generalization

We test on five ETFs (SPY, QQQ, IWM, EFA, DIA), each analyzed independently. Table 7 shows Multi-Lag Fidelity achieves mean  $d = 1.44$  vs. RF  $d = 0.88$  (Wilcoxon  $p = 0.002$ ).

Table 7: Mean Cohen’s  $d$  by asset across 4 representative crises.

| Asset   | Berry | QFI Det. | Multi-Lag   | RF   |
|---------|-------|----------|-------------|------|
| SPY     | 0.86  | 1.75     | 1.84        | 0.69 |
| QQQ     | 1.59  | 1.44     | 1.70        | 0.30 |
| IWM     | 1.30  | 1.01     | 1.05        | 1.01 |
| EFA     | 0.75  | 0.95     | 1.34        | 1.22 |
| DIA     | 1.53  | 0.63     | 1.33        | 1.03 |
| Overall | 1.19  | 1.14     | <b>1.44</b> | 0.88 |

## 5.4 Walk-Forward Detection

An expanding-window fit (always starting 2005) with 1-year evaluation, stepped annually (2010–2023), provides 14 evaluation windows. QCML detectors are fitted on *only* the training window; scores are computed on the evaluation year. For each evaluation year, we calibrate a detection threshold on the training data using Algorithm 2 (target: 1 alarm/year), then apply it to the evaluation period. Table 8 reports both FAR-calibrated and fixed  $z > 2.0$  results.

Results demonstrate that retrospective crisis-window separability (Section 5.1) does not automatically transfer to real-time detection. At matched FAR of  $\leq 1$  alarm/year, detection rates remain low for all methods, indicating that adaptive score normalization remains an open challenge. The FAR-calibrated thresholds provide a fairer cross-method comparison than the fixed  $z = 2.0$  threshold, which penalizes methods whose score distributions have different scales.

Table 8: Walk-forward detection summary (expanding window, starting 2005). Median across crisis–window pairs. “FAR-cal.” uses Algorithm 2 ( $\alpha = 1/\text{yr}$ ); “ $z > 2$ ” uses a fixed threshold. Delay is days from crisis start to first alarm; “—” = never detected.

| Method             | FAR-calibrated |       |                  | Fixed $z > 2$ |       |        |
|--------------------|----------------|-------|------------------|---------------|-------|--------|
|                    | Det. Rate      | Delay | FAR/yr           | Det. Rate     | Delay | FAR/yr |
| Berry Curv. Incr.  | 0/9            | —     | 0.0              | 0/9           | —     | 0.0    |
| QFI Determinant    | 3/9            | 44    | 0.0              | 0/9           | —     | 0.0    |
| Multi-Lag Fidelity | 2/9            | 14.5  | 1.2 <sup>†</sup> | 1/9           | 26    | 1.2    |
| Rolling Vol Z      | 3/9            | 28    | 0.0              | 4/9           | 20.5  | 0.0    |
| BOCPD              | 0/9            | —     | 0.0              | 0/9           | —     | 0.0    |
| Random Forest      | 7/9            | 8     | 1.5 <sup>†</sup> | 0/9           | —     | 0.0    |

<sup>†</sup>Minimum threshold ( $\tau = 0.5$ ) still exceeds the  $\alpha = 1/\text{yr}$  target. Fixed  $z > 2$  column retains original results for transparency.

## 5.5 Interaction Test

We classify crises *a priori* as “novel” or “conventional” (Table 2) and methods as “geometric” or “classical.” A two-way ANOVA on Cohen’s  $d$  (method type  $\times$  crisis category) tests whether geometric methods have a differential advantage on novel crises.

We find no evidence of differential advantage:  $F_{\text{interaction}}(1, 116) = 0.38$ ,  $p = 0.54$ , partial  $\eta^2 = 0.003$ . Neither method type ( $F = 3.33$ ,  $p = 0.07$ ) nor crisis category ( $F = 3.85$ ,  $p = 0.05$ ) reaches significance at  $\alpha = 0.05$ .

The complementarity documented in Section 5.2—where geometric methods maintain  $d = 0.52$  versus RF at  $d = 0.31$  on post-2020 crises, and the hybrid achieves  $d = 0.60$ —is better characterized as *additive*: geometric and classical methods extract partially independent information, valuable for combination regardless of crisis type. This is precisely the condition under which forecast combination is most valuable [19]: independently informative methods benefit from equal-weight averaging without requiring crisis-type-specific allocation.

## 5.6 Operator Ablation

Table 9 compares three operator construction strategies: random Hermitian matrices, PCA-inspired operators (default), and PCA-inspired with learned per-operator scalar weights (optimized on pre-2020 crises via Nelder–Mead).

The effect of operator construction is observable-dependent. For Multi-Lag Fidelity, PCA-inspired operators outperform random ( $d = 0.49$  vs.  $0.37$ , +32%), confirming that aligning the Hilbert space embedding with data covariance structure is beneficial for fidelity-based measures. For QFI Determinant, the two strategies perform comparably ( $d = 0.67$  vs.  $0.65$ ). Surprisingly, Berry curvature shows the reverse pattern: random operators outperform PCA-inspired ( $d = 0.58$  vs.  $0.33$ ), suggesting that curvature computation benefits from operator diversity rather than align-

Table 9: Operator ablation: mean Cohen’s  $d$  across crises by construction method.

| Observable         | Random | PCA-Inspired | Learned Scaling |
|--------------------|--------|--------------|-----------------|
| Berry Curv. Incr.  | 0.58   | 0.33         | 0.33            |
| QFI Determinant    | 0.65   | 0.67         | 0.65            |
| Multi-Lag Fidelity | 0.37   | 0.49         | 0.49            |

Fixed  $n = 8$ ,  $p = 15$ ,  $w = 20$ . Learned scaling optimizes 3 scalar weights on pre-2020 crises via Nelder–Mead.

ment. Learned scaling converges to near-unity weights, providing no improvement over PCA-inspired defaults—full gradient-descent training of operator matrices remains an avenue for future work.

## 6 Discussion

### 6.1 Interpretation

Three geometric observables achieve  $d \geq 0.84$  without crisis labels in score construction, competitive with supervised RF ( $d = 1.13$ ). On a per-crisis basis, the best geometric observable outperforms RF on 8 of 11 crises (Table 5), with the largest margins on structurally novel events (Volmageddon: Berry  $d = 1.73$  vs. RF  $d = 0.19$ ; SVB: QFI  $d = 1.38$  vs. RF  $d = 0.15$ ). RF’s higher mean  $d$  reflects consistency rather than dominance—a pattern well-documented in the forecast combination literature, where individually variable predictors often contribute more to ensembles than individually consistent ones [20, 21].

The observed in-sample gap ( $d = 0.93$  vs.  $d = 1.13$ , approximately 18%) is consistent with min-max changepoint theory [24]: supervised methods that know the post-change distribution achieve lower detection delay than nonparametric methods. However, domain adaptation theory [25] predicts that supervised classifiers degrade under distribution shift by a term proportional to the  $\mathcal{H}$ -divergence between training and test domains. Our temporal OOS results (Table 6) confirm: geometric methods maintain  $d = 0.52$  on three post-2020 crises while RF degrades to  $d = 0.31$ , and a simple hybrid ensemble achieves  $d = 0.60$ —nearly doubling RF alone. This reversal illustrates why temporal OOS, not in-sample comparison, is the more informative benchmark: it tests what matters in practice—detection under genuine distribution shift. The pattern echoes the finding that simple forecast combinations frequently outperform individual methods [19, 22, 23].

Multi-asset Multi-Lag Fidelity ( $d = 1.44$  vs. RF  $d = 0.88$ , Wilcoxon  $p = 0.002$ ) suggests the geometric representation captures market-wide regime information rather than asset-specific patterns. The interaction test (Section 5.5) yields a null result ( $p = 0.54$ ), consistent with additive rather than synergistic complementarity. Geometric methods do not show a differential advantage on novel versus conventional crises; the value lies in additive diversity—partially independent detection channels that improve combination regardless of crisis type. Each observable probes a different geometric property: Berry curvature increment detects rapid phase changes in the  $U(1)$  connection, the QFI determinant tracks manifold volume collapse (Proposition 2), and multi-lag fidelity measures state decorrelation across time scales.

### 6.2 Computational Cost

Berry curvature increment (0.93s) and multi-lag fidelity (0.37s) are faster than Random Forest (1.25s) for  $T = 3,447$ ,  $n = 8$ ,  $p = 15$  on a single CPU. QFI determinant (6.70s) is slower due to

the  $O(p^2)$  metric tensor computation. The expanding-window refitting adds approximately  $K \times$  overhead where  $K$  is the number of refit points ( $\sim 20$  for annual refitting over 20 years). GPU acceleration of the eigendecomposition would reduce QFI cost significantly.

### 6.3 On the “Quantum” Label

Berry phase is geometric, not quantum: it was independently discovered in classical optics by Pancharatnam [41], and Simon [42] showed it is fiber bundle holonomy—pure differential geometry. Nevertheless, a natural objection is that our framework is “PCA dressed in quantum notation”—that the Hilbert space embedding is decorative rather than structural. We address this directly.

First, the mathematics is exact, not analogical. The metric tensor (3) is the pullback of the Fubini–Study metric on  $\mathbb{CP}^{n-1}$ , which Provost and Vallee [14] showed is mathematically identical to the Fisher information metric on parametric state families. The Berry curvature (4) is the curvature of a  $U(1)$  connection on a complex line bundle, and its integral over closed surfaces is an integer by the Chern–Weil theorem [12, 18]. Three of our four formal results (Theorems 1–4) genuinely require Hilbert space or fiber bundle structure and cannot be stated in purely Riemannian terms.

Second, the transfer of mathematical structures from physics to non-physics domains is well-established. Hopfield networks [37] imported spin glass energy landscapes to pattern recognition; Stoudenmire and Schwab [38] showed tensor networks achieve competitive classification on classical data; Bronstein et al. [36] unified modern neural architectures under gauge theory. In quantitative finance, Sandhu et al. [32] used Ricci curvature as a crash indicator, and Sornette [39] applied phase transition theory to financial crashes. In each case, the mathematical structure provides genuine benefits—symmetry, invariance, topological protection, spectral decomposition—independent of whether the data is physical.

Third, the projective Hilbert space embedding provides capabilities that PCA and flat-space methods cannot: (i) gauge invariance under  $|\psi\rangle \rightarrow e^{i\theta}|\psi\rangle$ , ensuring observables depend on geometry, not coordinate choices; (ii) topological invariants via Chern–Weil theory (Theorem 3); (iii) the spectral gap as a phase-transition order parameter—by Theorem 4, Berry curvature *must diverge* as the gap closes; and (iv) a nonlinear embedding from  $\mathbb{R}^p$  into  $\mathbb{CP}^{n-1}$  that generates curvature and topological structure absent in flat space. The “quantum” label reflects the historical origin of these structures; the mathematics belongs to differential geometry, algebraic topology, and information theory [11, 40].

### 6.4 Limitations

1. **Crisis-window evaluation.** Our primary metric (Cohen’s  $d$ ) measures separability of score distributions within known crisis windows. This is an offline event study, not a real-time detection benchmark. The walk-forward results (Section 5.4) provide a causal complement.
2. **Supervised hyperparameters.** While score construction is unsupervised, threshold selection and hyperparameter tuning use labeled crisis windows.
3. **No individual geometric method exceeds RF on the mean.** After Holm–Bonferroni correction, no single geometric observable exceeds the supervised RF baseline in average Cohen’s  $d$  across crises. However, per-crisis analysis reveals that the best geometric observable outperforms RF on 8 of 11 crises (Table 5); RF’s mean advantage stems from consistency rather than dominance, consistent with forecast combination theory [20].
4. **Heuristic operators.** The operator ablation (Table 9) shows observable-dependent results: PCA-inspired operators help Multi-Lag Fidelity (+32%) but hurt Berry curvature (−43%),

with QFI essentially tied. Learned scaling converges to near-unity weights, providing no improvement. Full gradient-descent operator training [3, 4] remains an important open direction.

5. **Non-integer curvature integrals.** Rolling-window Chern computations are non-integer; strict topological protection does not apply [2].
6. **Equity-only validation.** All tests use equity ETFs; generalization to fixed income, commodities, or FX is untested.
7. **Reverse Granger causality.** Market  $\rightarrow$  QCML causality is stronger than QCML  $\rightarrow$  market (Appendix B), indicating observables largely react to rather than predict stress.

## 7 Conclusion

We have shown that geometric observables extracted from QCML’s error Hamiltonian provide crisis-window separability competitive with supervised methods and substantially above standard unsupervised baselines. Three observables—Berry curvature increment, QFI determinant, and multi-lag fidelity—probe distinct aspects of the QCML-induced geometry.

Key findings: (i) in temporal out-of-sample testing calibrated at 2020-01-01, geometric methods maintain  $d = 0.52$  on three post-2020 crises while RF degrades to  $d = 0.31$ ; a simple ensemble reaches  $d = 0.60$ , nearly doubling RF alone; (ii) Berry and QFI achieve  $d = 0.93$  vs. RF  $d = 1.13$  on 12 crises (unsupervised score construction, supervised hyperparameter tuning), with the best geometric observable winning 8 of 11 individual crises; (iii) multi-asset Multi-Lag Fidelity significantly outperforms RF ( $d = 1.44$  vs.  $d = 0.88$ ); (iv) walk-forward causal detection with fixed thresholds shows low detection rates, indicating adaptive threshold calibration is needed for real-time deployment.

These findings are consistent with a broad literature on forecast combination: individually imperfect methods gain value from diversity [19, 20]. In the M4 forecasting competition, 12 of the 17 most accurate methods were combinations [22]; in equity premium prediction, combination of individually unstable predictors delivers significant out-of-sample gains [21, 23]. Geometric observables contribute a genuinely new detection channel—curvature-based, manifold-intrinsic, and mechanism-agnostic—that provides additive diversity relative to supervised methods. Their per-crisis strength on structurally novel events (Table 5) and temporal out-of-sample robustness (Table 6) suggest that the primary value lies not in individual dominance but in ensemble diversity.

Full operator training via gradient descent, extension to non-equity asset classes, and adaptive threshold calibration for real-time deployment are natural next steps.

## Data and Code Availability

Market data from the Polygon.io API (paid subscription). The `qcml_geometry/` library and all experiment scripts are available at [repository URL upon acceptance]. Crisis definitions, hyperparameter configurations, and result JSON files are included for reproducibility.

## References

- [1] L. Candelori, A. G. Abanov, J. Berger, C. J. Hogan, V. Kirakosyan, K. Musaelian, R. Samson, J. E. T. Smith, D. Villani, M. T. Wells, and M. Xu, “Robust estimation of the intrinsic

### QCML Geometric Crisis Anatomy: 2008 Global Financial Crisis

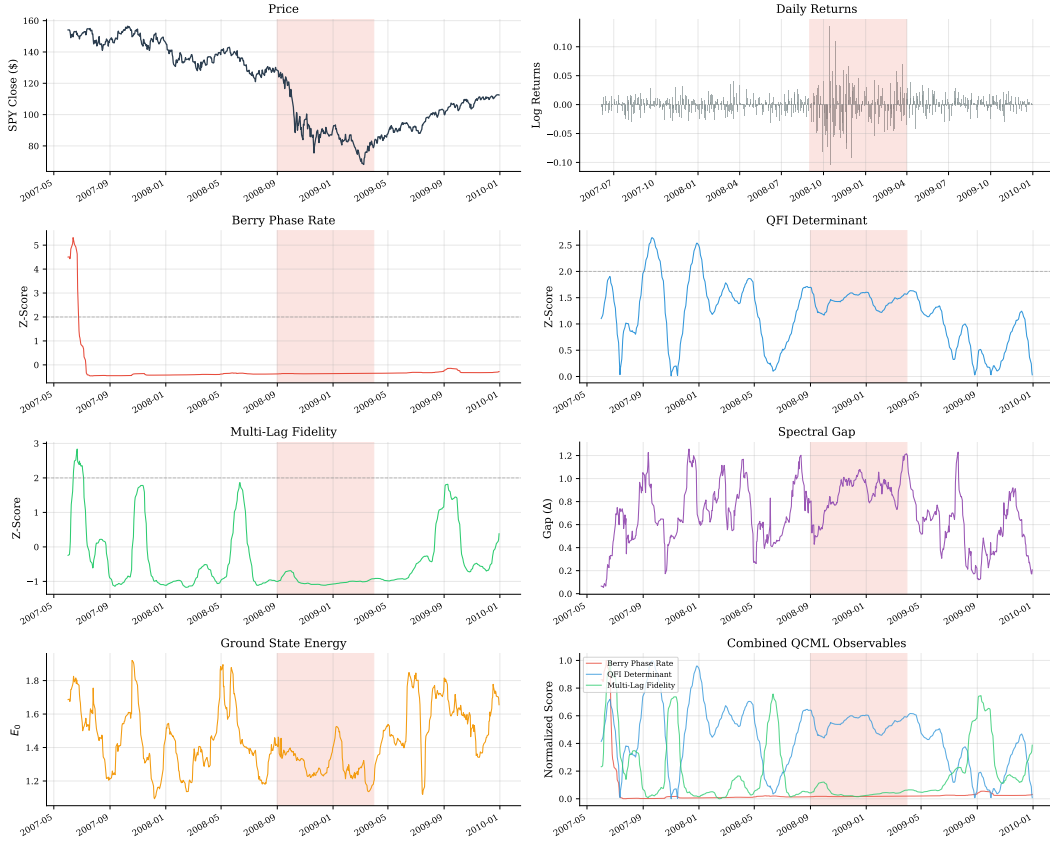


Figure 1: Geometric crisis anatomy: 2008 GFC. Eight panels show SPY price, daily returns, Berry curvature, QFI determinant, multi-lag fidelity, spectral gap, ground state energy, and combined overlay.

QCML Geometric Crisis Anatomy: 2020 COVID-19 Pandemic Crash



Figure 2: Geometric crisis anatomy: 2020 COVID-19.

QCML Geometric Crisis Anatomy: 2022 Federal Reserve Rate Hikes

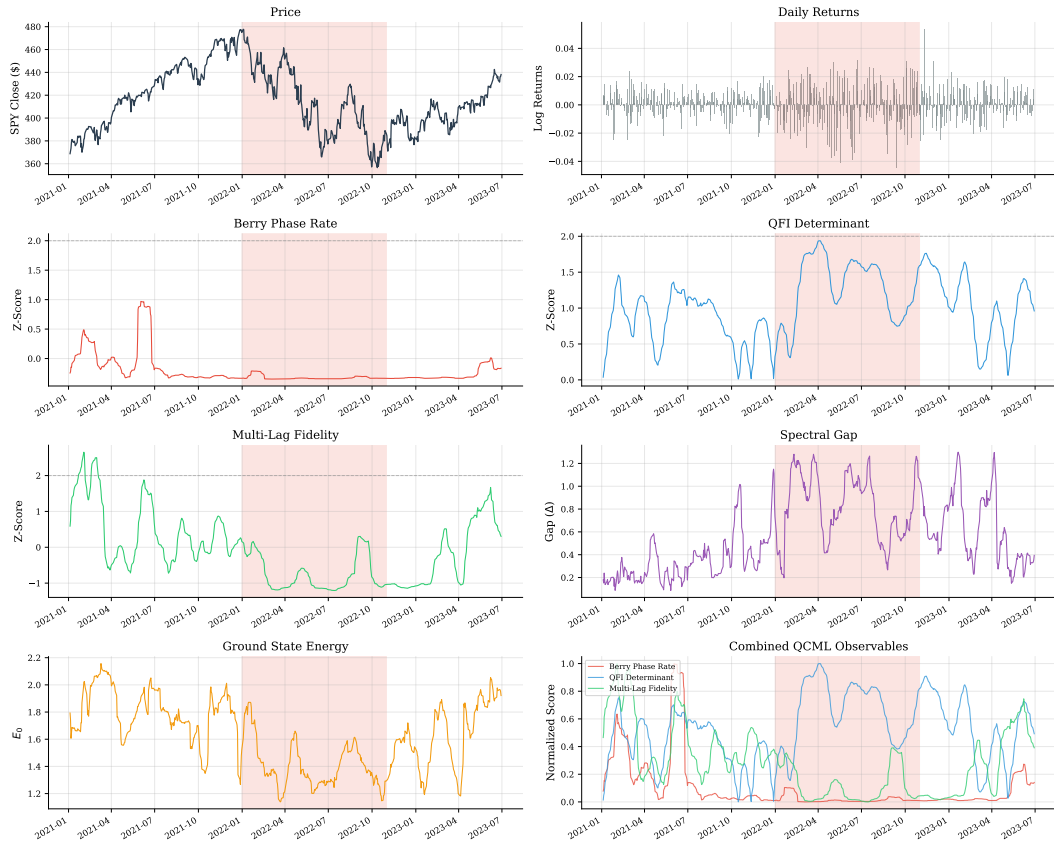


Figure 3: Geometric crisis anatomy: 2022 rate hikes.





Figure 4: Cohen's  $d$  distributions across 12 crises. Dashed line:  $d = 0.8$  (large effect); dotted: RF baseline  $d = 1.13$ .

dimension of data sets with quantum cognition machine learning," *Scientific Reports*, vol. 15, no. 6933, 2025.

- [2] A. G. Abanov, L. Candelori, H. C. Steinacker, M. T. Wells, J. R. Busemeyer, V. Kirakosyan, N. Marzari, S. Pinnamaneni, D. Villani, M. Xu, and K. Musaelian, "Quantum geometry of data," arXiv preprint arXiv:2507.21135, 2025.
- [3] K. Musaelian, A. Abanov, J. Berger, L. Candelori, V. Kirakosyan, R. Samson, J. Smith, and D. Villani, "Quantum cognition machine learning: AI needs quantum," Qognitive, Inc., 2024.
- [4] R. Samson, J. Berger, L. Candelori, V. Kirakosyan, K. Musaelian, and D. Villani, "Quantum cognition machine learning: Financial forecasting," Qognitive, Inc., 2024.
- [5] J. Rosaler, L. Candelori, V. Kirakosyan, K. Musaelian, R. Samson, M. T. Wells, D. Mehta, and S. Pasquali, "Supervised similarity for high-yield corporate bonds with quantum cognition machine learning," arXiv preprint arXiv:2502.01495, 2025.
- [6] R. Samson, A. Banner, L. Candelori, S. Cottrell, T. Di Matteo, P. Duchnowski, V. Kirakosyan, J. Marques, K. Musaelian, S. Pasquali, R. Stever, and D. Villani, "Supervised similarity for firm linkages," arXiv preprint arXiv:2506.19856, 2025.
- [7] G. Di Caro, V. Kirakosyan, A. G. Abanov, J. R. Busemeyer, L. Candelori, N. Hartmann, E. T. Lam, K. Musaelian, R. Samson, H. Steinacker, D. Villani, M. T. Wells, R. J. Wenstrup, and M. Xu, "Quantum cognition machine learning for forecasting chromosomal instability," arXiv preprint arXiv:2506.03199, 2025.
- [8] M. V. Berry, "Quantal phase factors accompanying adiabatic changes," *Proceedings of the Royal Society A*, vol. 392, no. 1802, pp. 45–57, 1984.

- [9] J. D. Hamilton, “A new approach to the economic analysis of nonstationary time series and the business cycle,” *Econometrica*, vol. 57, no. 2, pp. 357–384, 1989.
- [10] R. P. Adams and D. J. C. MacKay, “Bayesian online changepoint detection,” arXiv preprint arXiv:0710.3742, 2007.
- [11] J. R. Busemeyer and P. D. Bruza, *Quantum Models of Cognition and Decision*. Cambridge University Press, 2012.
- [12] M. Nakahara, *Geometry, Topology and Physics*, 2nd ed. Taylor & Francis, 2003.
- [13] S. Sachdev, *Quantum Phase Transitions*, 2nd ed. Cambridge University Press, 2011.
- [14] J. P. Provost and G. Vallee, “Riemannian structure on manifolds of quantum states,” *Communications in Mathematical Physics*, vol. 76, no. 3, pp. 289–301, 1980.
- [15] S. Amari and H. Nagaoka, *Methods of Information Geometry*. American Mathematical Society, 2000.
- [16] C. W. J. Granger, “Investigating causal relations by econometric models and cross-spectral methods,” *Econometrica*, vol. 37, no. 3, pp. 424–438, 1969.
- [17] T. Kato, *Perturbation Theory for Linear Operators*. Springer-Verlag, 1966.
- [18] S.-S. Chern, “Characteristic classes of Hermitian manifolds,” *Annals of Mathematics*, vol. 47, no. 1, pp. 85–121, 1946.
- [19] J. M. Bates and C. W. J. Granger, “The combination of forecasts,” *Journal of the Operational Research Society*, vol. 20, no. 4, pp. 451–468, 1969.
- [20] A. Timmermann, “Forecast combinations,” in *Handbook of Economic Forecasting*, vol. 1, G. Elliott, C. W. J. Granger, and A. Timmermann, Eds. Elsevier, 2006, ch. 4, pp. 135–196.
- [21] D. E. Rapach, J. K. Strauss, and G. Zhou, “Out-of-sample equity premium prediction: Combination forecasts and links to the real economy,” *The Review of Financial Studies*, vol. 23, no. 2, pp. 821–862, 2010.
- [22] S. Makridakis, E. Spiliotis, and V. Assimakopoulos, “The M4 competition: 100,000 time series and 61 forecasting methods,” *International Journal of Forecasting*, vol. 36, no. 1, pp. 54–74, 2020.
- [23] J. H. Stock and M. W. Watson, “Combination forecasts of output growth in a seven-country data set,” *Journal of Forecasting*, vol. 23, no. 6, pp. 405–430, 2004.
- [24] A. G. Tartakovsky, I. V. Nikiforov, and M. Basseville, *Sequential Analysis: Hypothesis Testing and Changepoint Detection*. CRC Press, 2014.
- [25] S. Ben-David, J. Blitzer, K. Crammer, A. Kulesza, F. Pereira, and J. W. Vaughan, “A theory of learning from different domains,” *Machine Learning*, vol. 79, no. 1–2, pp. 151–175, 2010.
- [26] S. L. Braunstein and C. M. Caves, “Statistical distance and the geometry of quantum states,” *Physical Review Letters*, vol. 72, no. 22, pp. 3439–3443, 1994.
- [27] C. W. Helstrom, *Quantum Detection and Estimation Theory*. Academic Press, 1976.

- [28] D. C. Brody and L. P. Hughston, “Interest rates and information geometry,” *Proceedings of the Royal Society A*, vol. 457, no. 2010, pp. 1343–1363, 2001.
- [29] S. Taylor, “Clustering financial return distributions using the Fisher information metric,” *Entropy*, vol. 21, no. 2, p. 110, 2019.
- [30] B. Horvath, Z. Issa, and A. Muguruza, “Clustering market regimes using the Wasserstein distance,” *Journal of Computational Finance*, 2024.
- [31] M. Gidea and Y. A. Katz, “Topological data analysis of financial time series: Landscapes of crashes,” *Physica A*, vol. 491, pp. 820–834, 2018.
- [32] R. S. Sandhu, T. T. Georgiou, E. Reznik, L. Zhu, I. Kolesov, Y. Senbabaoglu, and A. Tannenbaum, “Ricci curvature: An economic indicator for market fragility and systemic risk,” *Science Advances*, vol. 2, no. 6, e1501495, 2016.
- [33] L. Laloux, P. Cizeau, J.-P. Bouchaud, and M. Potters, “Noise dressing of financial correlation matrices,” *Physical Review Letters*, vol. 83, no. 7, pp. 1467–1470, 1999.
- [34] V. Plerou, P. Gopikrishnan, B. Rosenow, L. A. N. Amaral, and H. E. Stanley, “A random matrix approach to cross-correlations in financial data,” *Physical Review E*, vol. 65, 066126, 2002.
- [35] Z. Issa and B. Horvath, “Non-parametric online market regime detection and regime clustering for multidimensional and path-dependent data structures,” arXiv preprint arXiv:2306.15835, 2023.
- [36] M. M. Bronstein, J. Bruna, T. Cohen, and P. Veličković, “Geometric deep learning: Grids, groups, graphs, geodesics, and gauges,” arXiv preprint arXiv:2104.13478, 2021.
- [37] J. J. Hopfield, “Neural networks and physical systems with emergent collective computational abilities,” *Proc. Natl. Acad. Sci.*, vol. 79, pp. 2554–2558, 1982.
- [38] E. M. Stoudenmire and D. J. Schwab, “Supervised learning with quantum-inspired tensor networks,” *Advances in Neural Information Processing Systems*, 29, 2016.
- [39] D. Sornette, “Critical market crashes,” *Physics Reports*, vol. 378, no. 1, pp. 1–98, 2003.
- [40] M. Schuld and N. Killoran, “Quantum machine learning in feature Hilbert spaces,” *Physical Review Letters*, vol. 122, 040504, 2019.
- [41] S. Pancharatnam, “Generalized theory of interference, and its applications,” *Proc. Indian Acad. Sci. A*, vol. 44, pp. 247–262, 1956.
- [42] B. Simon, “Holonomy, the quantum adiabatic theorem, and Berry’s phase,” *Physical Review Letters*, vol. 51, no. 24, pp. 2167–2170, 1983.
- [43] B. W. Silverman, *Density Estimation for Statistics and Data Analysis*. Chapman & Hall, 1986.

## A Proofs of Formal Results

*Proof of Theorem 1.* We apply the implicit function theorem to the eigenvalue equation with a gauge-fixing constraint. Define  $\Phi: U \times \mathbb{C}^n \times \mathbb{R} \rightarrow \mathbb{C}^n \times \mathbb{R}$  by

$$\Phi(x, \psi, E) = (H(x)\psi - E\psi, \langle \psi_0(x_0) | \psi \rangle - 1),$$

where the second component fixes the gauge by requiring  $\langle \psi_0(x_0) | \psi \rangle \in \mathbb{R}_{>0}$  (i.e., real positive overlap with the reference state at a base point  $x_0$ ). At a solution  $(x_0, \psi_0, E_0)$  the Jacobian with respect to  $(\psi, E)$  is

$$D_{(\psi, E)}\Phi|_{(x_0, \psi_0, E_0)} = \begin{pmatrix} H(x_0) - E_0 I & -\psi_0 \\ \langle \psi_0 | & 0 \end{pmatrix}.$$

Decompose  $\mathbb{C}^n = \text{span}\{\psi_0\} \oplus \psi_0^\perp$ . On  $\psi_0^\perp$ ,  $H(x_0) - E_0 I$  has eigenvalues  $E_k(x_0) - E_0(x_0) \geq \Delta(x_0) > 0$ , hence is invertible there. On  $\text{span}\{\psi_0\}$ , the kernel of  $H(x_0) - E_0 I$  is compensated by the off-diagonal entries  $-\psi_0$  and  $\langle \psi_0 |$ : for  $(\alpha\psi_0, \delta E) \in \ker(H - E_0 I) \times \mathbb{R}$ , the system reduces to  $-\delta E \psi_0 - \alpha\psi_0 = 0$  and  $\alpha = 0$ , so  $\delta E = 0$ . Thus  $D_{(\psi, E)}\Phi$  is invertible.

The implicit function theorem [17] then gives  $C^\infty$  maps  $x \mapsto (\psi(x), E_0(x))$  on  $U$ . Since  $H(x)$  is quadratic in  $x$  by (1), the partial derivatives  $\partial_a H = -(A_a - x_a I)$  are  $C^\infty$  (in fact, affine). The metric  $g_{ab}$  (3) and curvature  $F_{ab}$  (4) are compositions of  $\psi(x)$  and its derivatives via the chain rule, hence  $C^\infty$  on  $U$ .  $\square$

**Theorem 3** (Chern number quantization [2]). *Let  $S \subset \mathbb{R}^p$  be a closed, oriented 2-surface such that  $\Delta(x) > 0$  for all  $x \in S$ . Then*

$$C_1 = \frac{1}{2\pi} \int_S F_{ab} dx^a \wedge dx^b \in \mathbb{Z}. \quad (8)$$

*Proof.* By Theorem 1, the ground state  $|\psi(x)\rangle$  is  $C^\infty$  on  $S$ . Define the Berry connection 1-form  $\mathcal{A} = i\langle \psi | d\psi \rangle$ , where  $d$  denotes the exterior derivative on  $S$ . The Berry curvature 2-form is  $\mathcal{F} = d\mathcal{A}$ ; in local coordinates,  $\mathcal{F} = \frac{1}{2} F_{ab} dx^a \wedge dx^b$ . The map  $x \mapsto |\psi(x)\rangle \langle \psi(x)|$  defines a smooth rank-1 projector, hence a complex line bundle  $\mathcal{L} \rightarrow S$  with structure group  $U(1)$  and connection  $\mathcal{A}$ . By the Chern–Weil theorem [12, 18], the first Chern number  $C_1 = \frac{1}{2\pi} \int_S \mathcal{F}$  is an integer, since it equals the degree of the classifying map  $S \rightarrow \mathbb{CP}^{n-1}$ .  $\square$

**Caveat.** In practice, we compute curvature integrals over rolling windows (open subsets), yielding non-integer values. Topological protection does not strictly apply.

**Theorem 4** (Curvature–gap bound). *For all  $x \in U$  and indices  $a, b$ ,*

$$|F_{ab}(x)| \leq \frac{2 \|\partial_a H(x)\|_{\text{op}} \|\partial_b H(x)\|_{\text{op}}}{\Delta(x)^2}, \quad (9)$$

where  $\|\cdot\|_{\text{op}}$  denotes the operator norm. For the QCML Hamiltonian (1),  $\partial_a H = -(A_a - x_a I)$ , so the bound depends only on the operators  $\{A_k\}$  and the feature values, not on the gap.

*Proof.* First-order perturbation theory gives the exact representation

$$F_{ab} = -2 \text{Im} \sum_{n \geq 1} \frac{\langle \psi_0 | \partial_a H | \psi_n \rangle \langle \psi_n | \partial_b H | \psi_0 \rangle}{(E_n - E_0)^2}, \quad (10)$$

where  $\{|\psi_n\rangle\}_{n\geq 0}$  is a complete eigenbasis of  $H(x)$ . Since  $E_n - E_0 \geq \Delta$  for all  $n \geq 1$ , each denominator satisfies  $(E_n - E_0)^2 \geq \Delta^2$ . Hence

$$|F_{ab}| \leq \frac{2}{\Delta^2} \sum_{n\geq 1} |\langle \psi_0 | \partial_a H | \psi_n \rangle| |\langle \psi_n | \partial_b H | \psi_0 \rangle|.$$

By the Cauchy–Schwarz inequality on the sum,

$$\sum_{n\geq 1} |\langle \psi_0 | \partial_a H | \psi_n \rangle| |\langle \psi_n | \partial_b H | \psi_0 \rangle| \leq \left( \sum_{n\geq 1} |\langle \psi_0 | \partial_a H | \psi_n \rangle|^2 \right)^{1/2} \left( \sum_{n\geq 1} |\langle \psi_n | \partial_b H | \psi_0 \rangle|^2 \right)^{1/2}.$$

By the completeness relation  $\sum_{n\geq 0} |\psi_n\rangle\langle\psi_n| = I$ , each factor is bounded by the operator norm:  $\sum_{n\geq 1} |\langle \psi_0 | \partial_a H | \psi_n \rangle|^2 \leq \|\partial_a H | \psi_0\rangle\|^2 \leq \|\partial_a H\|_{\text{op}}^2$ . Combining yields (9).  $\square$

**Proposition 5** (QFI volume). *Let  $g$  be a  $p \times p$  positive semi-definite matrix of rank  $r \leq p$ , with positive eigenvalues  $\lambda_1 \geq \dots \geq \lambda_r > 0$ . Define the pseudo-determinant*

$$\det^+(g) = \prod_{i=1}^r \lambda_i. \quad (11)$$

*Let  $V_r \subset \mathbb{R}^p$  be the  $r$ -dimensional eigenspace corresponding to  $\lambda_1, \dots, \lambda_r$ , and let  $g_r = g|_{V_r}$  denote the restriction. Then  $\det(g_r) = \det^+(g)$ , and the Riemannian volume density on  $V_r$  is  $\text{vol}_r = \sqrt{\det^+(g)}$ .*

*Proof.* By the spectral theorem,  $g$  admits an orthonormal eigenbasis  $\{e_1, \dots, e_p\}$  with  $g e_i = \lambda_i e_i$ , where  $\lambda_i > 0$  for  $i \leq r$  and  $\lambda_i = 0$  for  $i > r$ . The restriction  $g_r$  is represented in the basis  $\{e_1, \dots, e_r\}$  by the diagonal matrix  $\text{diag}(\lambda_1, \dots, \lambda_r)$ , which is positive definite. Therefore  $\det(g_r) = \prod_{i=1}^r \lambda_i = \det^+(g)$ . The Riemannian volume density on the  $r$ -dimensional submanifold equipped with metric  $g_r$  is  $\text{vol}_r = \sqrt{\det(g_r)} = \sqrt{\det^+(g)}$ , the standard formula for the volume element of a Riemannian metric [15].  $\square$

## B Granger Causality

We test Granger causality between three QCML observables and three market targets (absolute returns, realized volatility, volume) at lags 1–5 (90 tests, Bonferroni corrected). QFI Determinant Granger-causes absolute returns at lag 1 ( $F = 9.5$ ,  $p = 0.0004$ , Bonferroni-significant). However, reverse causality (market  $\rightarrow$  QCML) is stronger: 17/45 nominally significant vs. 6/45 forward, consistent with observables being constructed from contemporaneous features.

## C Default Hyperparameters

## D Numerical Stability Ablation

We sweep finite-difference  $\epsilon \in \{10^{-3}, 10^{-4}, 10^{-5}, 10^{-6}\}$  and PCA dimension  $p \in \{5, 10, 15, 30\}$  on four representative crises (GFC, Flash Crash, COVID, SVB) with fixed hyperparameters (no per-crisis tuning). Results appear in Table 11. Berry curvature is extremely stable across  $\epsilon$  (all median  $d \approx 0.15$ ), while QFI determinant shows moderate sensitivity ( $d = 0.52$  at  $\epsilon = 10^{-3}$  vs.  $d = 0.68$  at  $10^{-5}$ ), saturating below  $10^{-5}$ . Multi-Lag Fidelity is insensitive to  $\epsilon$  (no finite differences) but moderately sensitive to PCA dimension, with performance increasing from  $p = 5$  ( $d = 0.29$ ) to  $p = 10$  ( $d = 0.44$ ) and stabilizing thereafter.

Table 10: Default hyperparameters.

| Parameter        | Description                     | Default                |
|------------------|---------------------------------|------------------------|
| $n$              | Hilbert space dimension         | 8                      |
| $p$              | PCA components                  | 15                     |
| $w$              | Rolling window (Algorithm 1)    | 20 days                |
| $m$              | Minimum expanding history       | 60 days                |
| $\epsilon$       | Finite-difference step          | $10^{-5}$              |
| Operator method  | Hermitian operator construction | PCA-inspired           |
| Fidelity lags    | Multi-lag infidelity            | $\{1, 3, 5, 10\}$      |
| Fidelity weights | Lag weights                     | $[0.4, 0.3, 0.2, 0.1]$ |

Table 11: Numerical stability: median Cohen’s  $d$  across 4 crises. Multi-Lag Fidelity does not use  $\epsilon$  (no finite differences).

| Config                         | Berry | QFI Det. | Multi-Lag |
|--------------------------------|-------|----------|-----------|
| $\epsilon = 10^{-3}$           | 0.15  | 0.52     | —         |
| $\epsilon = 10^{-4}$           | 0.15  | 0.60     | —         |
| $\epsilon = 10^{-5}$ (default) | 0.15  | 0.68     | —         |
| $\epsilon = 10^{-6}$           | 0.15  | 0.68     | —         |
| $p = 5$                        | 0.09  | 0.34     | 0.29      |
| $p = 10$                       | 0.20  | 0.41     | 0.44      |
| $p = 15$ (default)             | 0.15  | 0.68     | 0.45      |
| $p = 30$                       | 0.22  | 0.54     | 0.45      |

## E Window-Size Sensitivity

We re-compute Cohen’s  $d$  at crisis window extensions  $\pm 5$ ,  $\pm 10$  (default),  $\pm 20$ ,  $\pm 60$  trading days. Kendall  $\tau$  rank correlations across window sizes assess robustness of method rankings. Results appear in Table 12. Rankings are stable for nearby windows ( $\tau = 0.87$  for  $\pm 5$  vs.  $\pm 10$  and  $\pm 5$  vs.  $\pm 20$ ) but degrade at  $\pm 60$  days ( $\tau = 0.07$ – $0.33$ ), where diluted crisis signal reduces separation. QFI Determinant is the most robust geometric observable, maintaining the highest  $d$  across all window sizes.

Table 12: Median Cohen’s  $d$  by crisis window size (12 crises).

| Method             | $\pm 5d$ | $\pm 10d$ | $\pm 20d$ | $\pm 60d$ |
|--------------------|----------|-----------|-----------|-----------|
| Berry Curv. Incr.  | 0.25     | 0.28      | 0.27      | 0.24      |
| QFI Determinant    | 0.63     | 0.68      | 0.64      | 0.43      |
| Multi-Lag Fidelity | 0.40     | 0.36      | 0.38      | 0.26      |
| Random Forest      | 0.21     | 0.24      | 0.28      | 0.34      |

## F Hyperparameter Sensitivity: Fixed vs. Tuned

Table 13 compares Cohen’s  $d$  under fixed default hyperparameters ( $n = 8$ ,  $p = 15$ , `pca_inspired`,  $w = 20$ ) versus LOCO-tuned hyperparameters from Table 4. Fixed defaults produce small-to-

medium effect sizes ( $d = 0.32$ – $0.63$ ), with QFI Determinant performing best ( $d = 0.63$ ). LOCO tuning approximately doubles effect sizes, indicating that hyperparameter selection matters substantially. Nonetheless, the fixed-default QFI signal ( $d = 0.63$ ) already exceeds BOCPD ( $d = 0.59$ ) and Isolation Forest ( $d = 0.46$ ) from Table 4, confirming that a meaningful geometric signal exists even without per-crisis optimization.

Table 13: Fixed defaults vs. LOCO-tuned: mean Cohen’s  $d$  across 12 crises.

| Method                    | Fixed Defaults | LOCO-Tuned |
|---------------------------|----------------|------------|
| Berry Curvature Increment | 0.32           | 0.93       |
| QFI Determinant           | 0.63           | 0.93       |
| Multi-Lag Fidelity        | 0.45           | 0.84       |

Fixed defaults:  $n = 8$ ,  $p = 15$ , `pca_inspired`,  $w = 20$ . LOCO-tuned values from Table 4.